Xiangling Xu 许湘灵

PhD student, PhIQuS, Inria Saclay

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My Website

Google Scholar

ORCID



Research Interests

I am interested in the capabilities and limits of quantum physics. To explore these questions, I study correlations in quantum networks, certifiable properties of quantum many-body systems, and cryptographic approaches to testing quantum devices. My main mathematical tools are the theory of operator algebras and noncommutative polynomial optimization.

Education

PhD in Computer Science, Institut Polytechnique de Paris 2023 – Present

Project: *Quantum correlations in causal structures: characterization and application.*
Expected completion: 2027-05; supervisor: Marc-Olivier Renou.

MSc in Mathematics, ETH Zürich – Institute for Theoretical Physics 2020 – 2023
Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective.*
Supervisor: Renato Renner; GPA: 5.69/6.00.

HBSc in Mathematics and Physics (Specialist), University of Toronto 2016 – 2020
With High Distinction; CGPA: 3.95/4.00.

Publications and Preprints

1. **An operational characterization of finite-dimensional quantum theory**.
   L. J. Dmello, X. Xu, M.-O. Renou, and D. Gross.
   2026-10. arXiv: 2610.03694 [quant-ph].

2. **Quantum inflation and source-transfer hierarchies are complete for tree networks**.
   X. Xu, I. Klep, and M.-O. Renou.
   2026-09. arXiv: 2609.19310 [quant-ph].

3. **Fermions are fundamentally more nonlocal than Bosons**.
   F. M. Kalarde, S. Boreiri, X. Xu, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou.
   2026-06. arXiv: 2606.12363 [quant-ph].

4. **The bulk spectral gap is certifiable from above but uncomputable from below**.
   X. Xu, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou.
   2026-06. arXiv: 2606.03836 [quant-ph].

5. **Comment on "Quantum theory based on real numbers cannot be experimentally falsified": on the compatibility of physical principles with information theory for fermions**.
   F. M. Kalarde, X. Xu, and M.-O. Renou.
   2026-04. arXiv: 2604.07425 [quant-ph].

6 **Two convergent NPA-like hierarchies for the quantum bilocal scenario**.
M.-O. Renou, X. Xu, and L. T. Ligthart.
Journal of Mathematical Physics, vol. 67, no. 1, Art. no. 012203, 2026-01. DOI: 10.1063/5.0211008

7 **Quantitative quantum soundness for all multipartite compiled nonlocal games**.
M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and X. Xu.
2025-09. arXiv: 2509.25145 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

8 **Quantitative quantum soundness for bipartite compiled Bell games via the sequential NPA hierarchy**.
I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, X. Xu, and Y. Zhao.
2025-07. arXiv: 2507.17006 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

9 **Quantitative Tsirelson's theorems via approximate Schur's lemma and probabilistic Stampfli's theorems**.
X. Xu, M.-O. Renou, and I. Klep.
2025-05. arXiv: 2505.22309 [quant-ph].

10 **On the kinematic morphology around haloes**.
X. Luo, X. Xu, and X. Wang.
Monthly Notices of the Royal Astronomical Society, vol. 518, no. 4, pp. 6059–6064, 2023. DOI: 10.1093/mnras/stac3500

11 **Quantum nonlocality in bilocal networks: an operator algebraic perspective**.
M.S. thesis, ETH Zurich, 2023. DOI: 10.3929/ethz-b-000613670

Selected Talks

My work has been presented at leading international conferences in quantum information (QIP), quantum cryptography (QCrypt), optimization (SIAM), and operator theory (IWOTA), and in selected contributed talks at GDR TeQ, a major national meeting of France's CNRS Quantum Technologies network.

The bulk spectral gap is RE-complete: certifiable from above, uncomputable from below

2026-11 Journées Informatique Quantique, Toulouse, France (contributed).
Gif-sur-Yvette, France (contributed).

Quantitative quantum soundness for compiled nonlocal games and the sequential NPA hierarchy

2026-08 QCrypt, Ottawa, Canada (contributed).
2026-06 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom (invited).
2026-01 QIP 2026, Riga, Latvia (two contributed talks: bipartite and multipartite games).
2025-11 3rd Colloquium GDR TeQ, Grenoble, France (contributed).
2025-10 YQIS25, Barcelona, Spain (contributed).
2025-07 IWOTA, Enschede, Netherlands (invited).
2025-06 IQC-PCQT-Quantum Saclay Workshop, Paris, France (invited).

Characterizing quantum bilocal networks with generalized NPA hierarchies

2024-09 CausalWorlds, Waterloo, Canada (contributed).
2023-11 Defi EQIP Workshop, Lyon, France (contributed).
2023-10 Inria Saclay (Saclay Quantum Seminar, invited).

Selected Talks (continued)

Other research seminars

2026-11 University of Cologne (David Gross).
2026-09 Télécom Paris (Quriosity).
2026-08 Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area (Ranyiliu Chen).
2026-07 Nanjing University (Penghui Yao).
2026-02 Institut de physique théorique, CEA (Jean-Daniel Bancal).
2025-12 Quantum Meets (online, invited).
AMSS, Chinese Academy of Sciences (Jie Wang).
2025-11 EPFL (Thomas Vidick).
2025-10 Freie Universität Berlin (Jens Eisert).
2025-02 ENS Paris (QAT team).
2024-07 University of Waterloo (William Slofstra).
2023-12 University of Cologne (David Gross).

Posters

2026 CQCC & GQSF, Shenzhen, China.
2024 2nd Colloquium GDR TeQ, Paris, France; YQIS24, Paris, France.

Research Visits

David Gross — University of Cologne. 2026-10 – 2026-11

Penghui Yao — Nanjing University. 2026-07

Jie Wang — AMSS, Chinese Academy of Sciences. 2026-05, 2025-12

Thomas Vidick — EPFL. 2025-10 – 2025-11

Igor Klep — University of Ljubljana. 2024-09

William Slofstra — Institute for Quantum Computing, University of Waterloo. 2024-07

David Gross — University of Cologne. 2023-12

Academic Service

Organization

Co-organizer, COMPUTE (QuantERA) 2024 – Present
Five-partner European collaboration on quantum networks and noncommutative polynomial optimization (Inria Saclay, LAAS-CNRS Toulouse, TU Darmstadt, University of Ljubljana, ICFO; EUR 0.99 million total project funding).

- Contributed to the successful QuantERA proposal and coordinated consortium meetings and project dissemination.

Co-organizer, Gaspard Monge Visiting Professorship and seminar series 2025 – 2026
Visiting professor Igor Klep; seminars on polynomial optimization and quantum theory, École Polytechnique and Inria Saclay.

- Helped draft the funding application and organized invited talks by Jean-Bernard Lasserre, Monique Laurent, Antonio Acín, Hamza Fawzi, William Slofstra, and Andreas Bluhm.

Co-organizer, 828th WE-Heraeus Seminar 2025
Operator Theory and Polynomial Optimization in Quantum Information Theory, Physikzentrum Bad Honnef.

· Contributed to scientific programme development, speaker invitations, and overall organization.

Co-organizer, Saclay Quantum Workshop 2023
One-day workshop on Navascués–Pironio–Acín (NPA) hierarchies.

· Contributed to programme design, speaker invitations, and the organization of tutorials and research talks.

Peer Review

Referee: *Quantum* (journal), *Operator Theory* (Springer reference work; chapter), QIP, CausalWorlds.

Selected Awards and Funding

Erasmus+ Mobility Funding (EUR 1,154) 2026
Research visit to the University of Cologne.

Erasmus+ Mobility Funding (EUR 1,274) 2024
Research visit to the University of Ljubljana.

Class of 3To and Associates Scholarship in Mathematics and Physics (CAD 2,050) 2018
University of Toronto · Selective merit scholarship for outstanding academic performance in second- or third-year Mathematics and Physics.

International Experience (IE) Award (CAD 2,500) 2018
University of Toronto · Summer research exchange at the University of Liverpool.

Summer Undergraduate Research Program (SURP) (CAD 9,000) 2017
Canadian Institute for Theoretical Astrophysics (CITA) · Selected as the only first-year student in the 2017 cohort of a 16-week paid research program recruiting undergraduates internationally. The project led to a publication.

President's Scholar of Excellence (CAD 10,000) 2016
University of Toronto · One of approximately 50 university-wide recipients (over 43,000 first-entry undergraduate admissions in 2016), with additional benefits such as faculty mentorship.

Teaching Experience

2024, 2025 **Guest lecturer** (one lecture per year), *Recent Trends in Parallel, Distributed, and Quantum Computing*, Quantum and Distributed Computer Science (QDCS) Master's program, Université Paris-Saclay.

2022 – 2023 **Teaching assistant**, *Linear Algebra I & II*, ETH Zürich.

2021 – 2022 **Teaching assistant**, *Analysis I & II*, ETH Zürich.

Skills

Computing Python, MATLAB, C, Mathematica; ncpol2sdpa, YALMIP.

Languages Chinese (native), English (fluent), German (beginner), French (beginner).